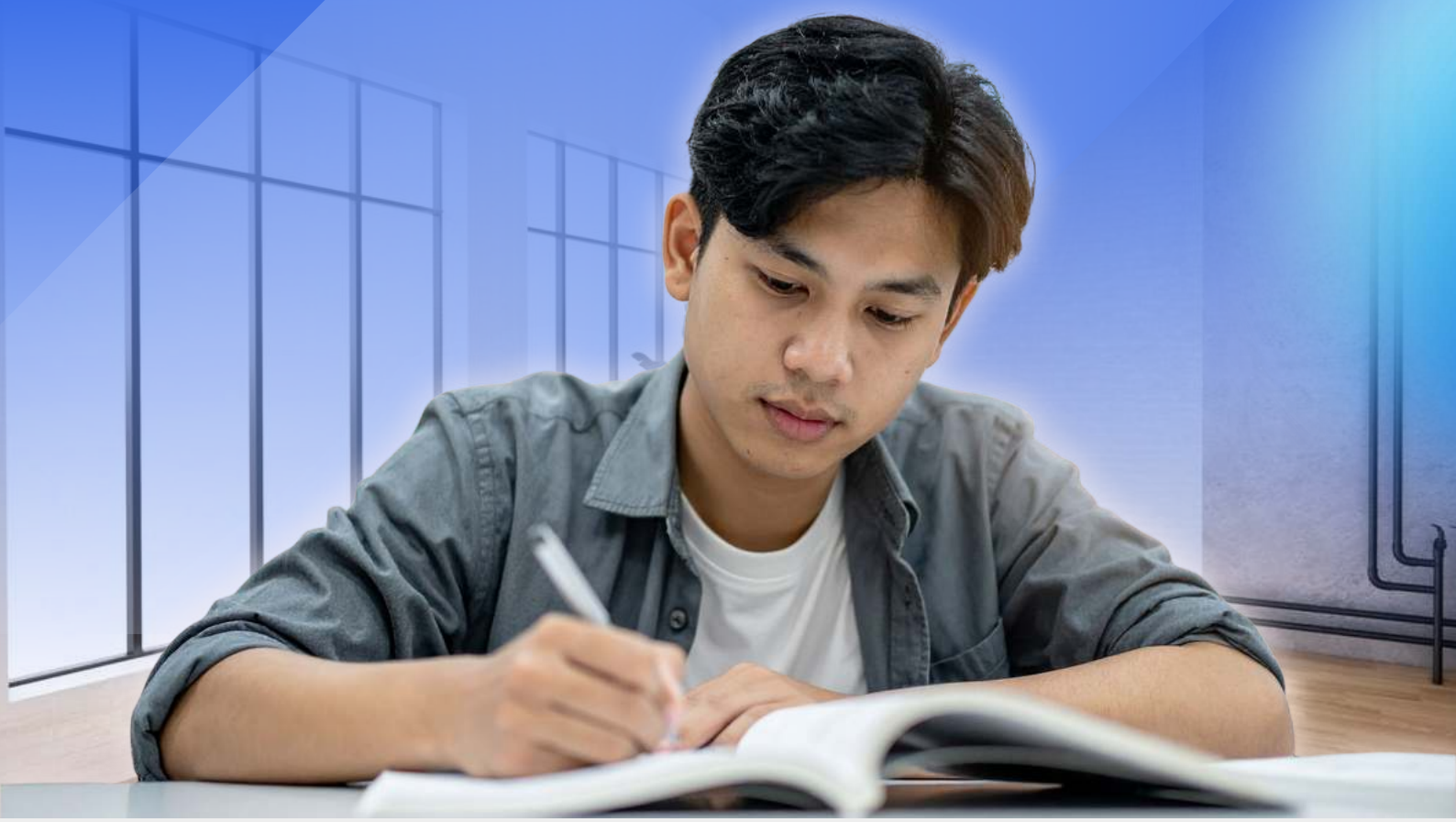
Lenterago

TRY OUT #8

SAT-Math

61008

44 QUESTIONS



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MODUL 1 & 2

22 QUESTIONS

DIRECTIONS

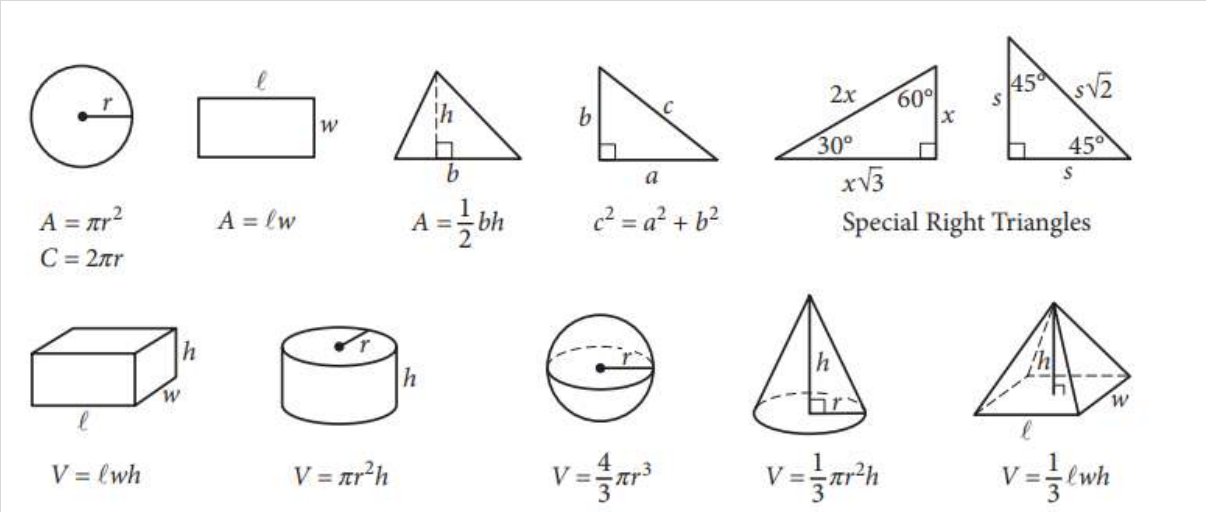
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function *f* is the set of all real numbers *x* for which *f(x)* is a real number

REFERENCE



The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2p.

The sum of the measures in degrees of the angles of a triangle is 180.

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Module 1 & 2

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find more than one correct answer, write and circle only one answer.
- Your answer can be up to 5 characters for a positive answer and up to 6 characters (including the negative sign) for a negative answer, but no more.
- If your answer is a fraction that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a decimal that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a mixed number (such as 3 1/2), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include symbols such as a percent sign, comma, or dollar sign in your circled answer.

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NO	MATERI	SUB MATERI
1	Algebra	Linear Equation 1 Variable
	If $2x + 3 = 9$ what is the value of $6x - 9$?	
2	Algebra	Word Problem
	A bag of Nut Mix contains x ounces of walnuts, 15 ounces of peanut and 20 ounces of pecans. Which of the following expresses the fraction of the mix, by weight, that is walnuts? A. $\frac{x}{35}$ B. $\frac{x}{35-x}$ C. $\frac{x}{x+35}$ D. $\frac{35-x}{35+x}$	
3	Algebra	Word problem Linear Equation
	The cost in dollar, C, to manufacture n necklaces is given by equation $C(n) = an + b$, where a and b are positive constants. In this equation, what does a represent? A. the fixed cost , in dollars, independent of any necklaces being manufactured B. the total cost , in dollars, to produce n necklaces, not including fixed cost C. the total cost , in dollars, to produce n necklaces, including fixed cost D. the cost , in dollars, to produce one necklaces, not including any fixed cost	
4	Algebra	Linear Equation
	$hx + 4y =- 3$ The equation above is the equation of a line in the xy-plane and h is a constant. If the slope of this line is $- 13$, what is the value of h?	
5	Algebra	Word Problem Linear Equation
	The cost of renting a campsite for up to 10 days is $300 for first day and $240 for each additional day. Which of the following equation give the cost $f(x)$, in dollars, of renting the campsite for x days, where x is a positive integer and less than or equal to 10? • $f(x) = 300x + 240$ • $f(x) = 300x - 60$ • $f(x) = 240x + 60$ • $f(x) = 240x + 300$	

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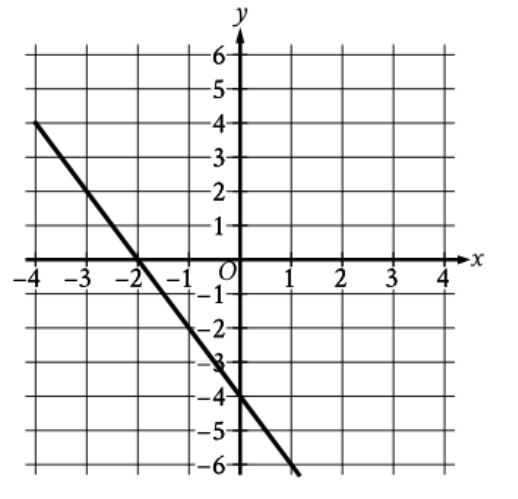
6	Algebra	System of Linear Equation

One of the two equations in a linear system is $4x + 12y = 20$. The system has no solution. Which of the following could be the other equation in the system?

A. $x + 3y = 5$
B. $x + 3y =- 20$
C. $6x - 2y = 0$
D. $6x + 2y = 10$

7	Algebra	Linear Function

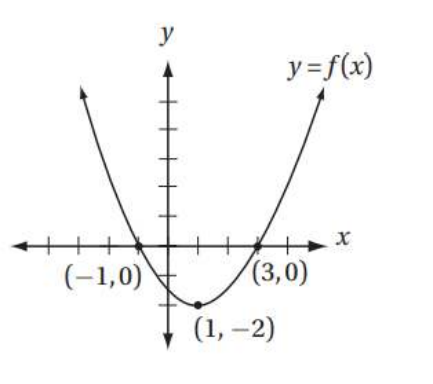
The graph of $y = f(x) + 6$ is shown as below



Which equation define linear function f ?

A. $f(x) =- 2x - 12$
B. $f(x) =- 2x - 4$
C. $f(x) =- 2x - 10$
D. $f(x) =- 2x + 2$

8	Advanced Math	Function and Transformation



The graph shown is the graph of the quadratic function $y = f(x)$.
A function g is defined by the equation $g(x) = f(x) + b$. If the equation $g(x) = 0$ has exactly one solution, what is the value of b?

A. -2
B. -1
C. 1
D. 2

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9	Advanced Math	Exponential

If $\left(6^{-2}\right)\left(m^{-2}\right) = \frac{1}{16}$ what is the value of m^2?

A. $\frac{1}{9}$
B. $\frac{4}{9}$
C. $\frac{9}{16}$
D. $\frac{9}{4}$

10	Advanced Math	Exponential Function

$$f(x) = 5492(0.64)^{\frac{x}{12}}$$

The function f gives the value , in dollars, of a certain piece of equipment after x months of use. If the value of equipment decreased each year by $p\%$ of its value the preceding year, what is the value of p?

A. 4
B. 5
C. 36
D. 64

11	Advanced Math	Rational Expression

The equation $\frac{12x^2+6x-23}{bx-3} =- 2x - \frac{23}{bx-3}$ is true for all value of $x \neq \frac{3}{b}$ where b is constant. What is the value of b?

A. -12
B. -6
C. 4
D. 6

12	Advanced Math	Quadratic Equation

A ball is kicked from a platform. The equation $h =- 5.4t^2 + 7t + 2$ represents this situation, where h is the height of the object above the ground, in meters, t second after it is kicked. Which number represents the height, in meters, from which the object was kicked?

A. 0
B. 5,4
C. 7
D. 2

13	Advanced Math	System of Nonlinear Equation

$$y = x^2 - 4x + 4$$
$$y = 4 - x$$

If the ordered pair (x, y) satisfied the system of equation above , what is the possible positive value of x?

14	Advanced Math	Exponential + Nested Function

$$f(x) = 304(2)^x$$

The function f is defined by the given equation. If $h(x) = f(x - 4)$, which of the following equation define h?

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A. $h(x) = 19(2)^x$
B. $h(x) = 68(2)^x$
C. $h(x) = 304(16)^x$
D. $h(x) = 19(4)^x$

15 | Problem Solving and Data Analysis | Percentages

An online trading company charges a 3% commission for all stock purchases. If a trader purchases 200 shares of a stock through this company and charged $\$3,399$ including commission, what is the cost per share for this stock?

A. \$16.45
B. \$16.48
C. \$16.50
D. \$16.52

16 | Problem Solving and Data Analysis | Probability

For a science project, Hana recorded whether it rain each weekday and weekend for 12 weeks, The result is summarize in the table below

Weekday and Weekend Day Rain for 12 Weeks

	Rain	No rain	Total
Number of weekdays	12	48	60
Number of weekend days	8	16	24
Total	20	64	84

If one of the day which there was no rain is selected at random, what is the probability the day was weekday?

A. $\frac{4}{21}$
B. $\frac{1}{4}$
C. $\frac{2}{3}$
D. $\frac{3}{4}$

17 | Problem Solving and Data Analysis | Percentage

Year	Subscriptions
2024	5,600
2025	5,880

The manager of online magazine received the report above on the number of subscription to the platform. The manager estimated that the percent increase from 2024 to 2025 is would be double the percent increase from 2025 to 2026 due to the recent reading trend. How many subscription did the manager expect would be sold in 2026?

A. 6,020
B. 6,027
C. 6,440
D. 6,468

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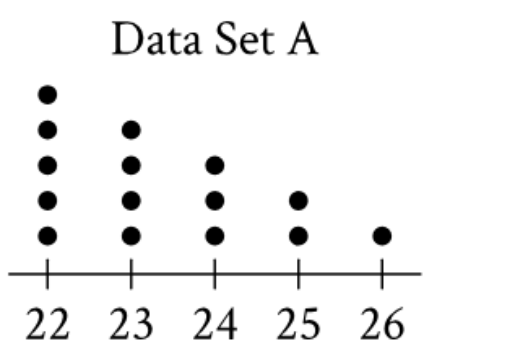
18	Problem Solving and Data Analysis	Comparing Data



The dot plot represents the 15 value of data set A. Data set B is created by multiplying 5 to each value of data set A.
Which of the following correctly compares the mean and standard deviation of data set A and B?

A. The mean of data set B is equal to data set A and the standard deviation of data set B is 5 time of data set A
B. The mean of data set B is equal to data set A and the standard deviation of data set B is also equal to data set A
C. The mean of data set B is 5 times of data set A and the standard deviation of data set B is 5 time of data set A
D. The mean of data set B is a fifth of set A and the standard deviation of data set B is 5 time of data set A

19	Problem Solving and Data Analysis	Probability

A jar contains only red, white and blue marbles. It contains twice as many red marbles as white marbles and three times as many white marbles as blue marbles. If a marble is chosen at random, what is the probability that it is not red?

A. $\frac{1}{5}$
B. $\frac{2}{5}$
C. $\frac{3}{5}$
D. $\frac{4}{5}$

20	Geometry	Area of similar figure

Square A has side length that 100 times the side length of square B. The area of square A is k time the area of square B. What is the value of k?

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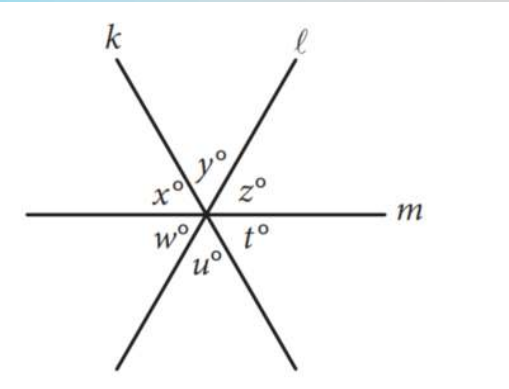
21	Geometry	Line and Angle

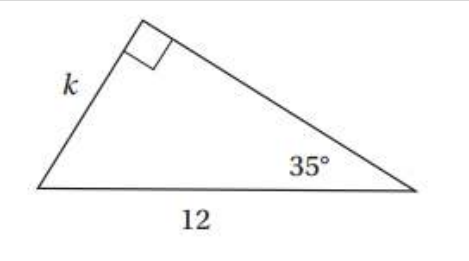


In the figure above, lines k, l and m intersect at a point. If $x + y = u + w$ which of the following must be true?

I. $x = z$
II. $y = w$
III. $z = t$

A. I and II only
B. I and III only
C. II and III only
D. I, II, and III

22	Geometry	Trigonometry



In the triangle above, what is the value of k?
$(sin\ 35° = 0,574, cos\ 35° = 0,819, tan\ 35° = 0,700)$

A. 6.00
B. 6.88
C. 8.40
D. 9.83

23	Algebra	Inequality

Which of the following ordered pairs (x, y) satisfies the inequality $5x - 3y < 4$?

I. (1,1)
II. (2,5)
III. (3,2)

A. I only
B. II only

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C. I and II only
D. I and III only

24 | Algebra | Rate Problems

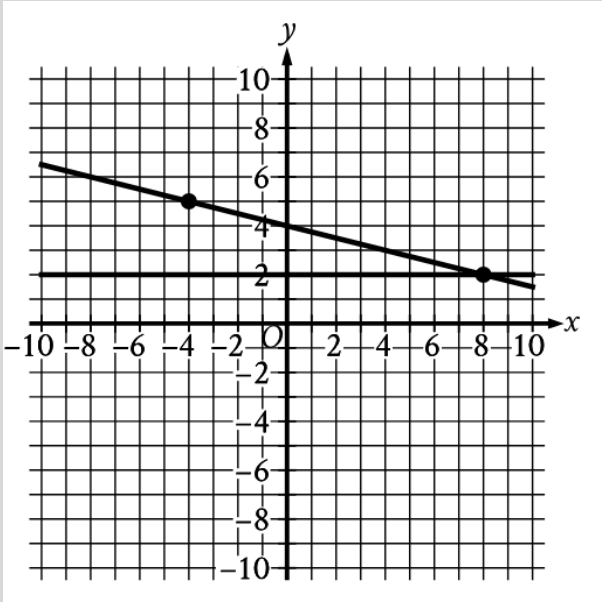
A kayaker can only travel between 8 miles per hours and 12 miles per hours. What is the minimum hours it could take the kayaker to travel 132 miles?

25 | Algebra

$$-x - wy = -337$$
$$2x - wy = 47$$

In the given system of equation , w is a constant. And it has solution at $(q, 19)$, where q is a constant. What is the value of w?

26 | Algebra | System of Linear Equation



If a new graph of three linear equation is created by using the system of equation shown and the equation $3x - 4y = 16$, how many solutions (x, y) will be the resulting system of the three equations have?

A. Zero
B. Exactly one
C. Exactly two
D. Infinitely many

27 | Algebra | Word problem of Linear Function

$$C = 50h + 125$$

The equation above gives the amount C, in dollar, a house keeper charge for a job that take h hours. Ms. Sanchez and Mr. Roland each hired this house keeper to clean their house. The house keeper work 3 hours longer on Ms. Sanchez house. How much more did the house keeper charge Ms. Sanchez than Mr. Roland?

A. $125
B. $150
C. $225
D. $275

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28	Algebra	System of Inequality

A big baker house is sponsoring a bake sale that sells cookies and brownies. Each cookie cost \$1.50, and each brownies cost \$2.25. The sponsor goals for the day to sell at least 55 items and to earn at lest \$100 of revenue. Let x be the number of cookies sold and y be the number if brownies sold. Which of the following system of inequalities represent the sponsor goals?

A. $x + y \leq 100$
$1.5x + 2.25y \leq 55$

B. $x + y \geq 100$
$1.5x + 2.25y \geq 55$

C. $x + y \geq 55$
$1.5x + 2.25y \leq 100$

D. $x + y \geq 55$
$1.5x + 2.25y \geq 100$

29	Algebra	System of Equation

$$2ax - 15 = 3(x + 5) + 5(x - 1)$$

In the equation above, a is a constant. If no value if x satisfies the equation, what is the value of a?

A. 1
B. 2
C. 4
D. 8

30	Algebra	Linear Function

n	1	2	3	4
$f(n)$	8	5	2	-1

The table above show some value of the linear function f. Which of the following define f?

A. $f(n) = -3n - 8$
B. $f(n) = 2n - 11$
C. $f(n) = -3n + 11$
D. $f(n) = 3n + 11$

31	Advanced Math	Quadratic Equation

$$x + \frac{36}{x} = 12$$

If $x > 0$, what is the solution to the equation above?

32	Advanced Math	Exponential

If $4x - 2y = 20$, what is the value of $\frac{81^x}{9^y}$?

A. 9^5
B. 9^{10}
C. 3^{10}
D. 3^5

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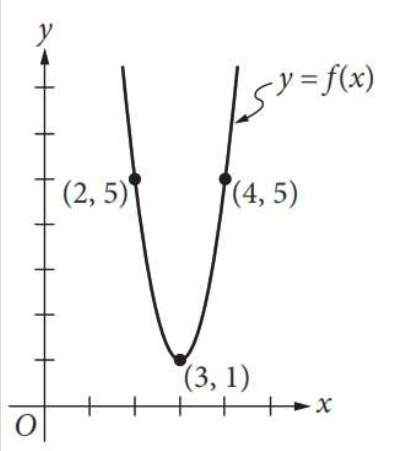
33	Advanced Math	Radical Equation	
	If $a = 4\sqrt{2}$ and $2a = \sqrt{2b}$, what is the value of b?		
34	Advanced Math	Polynomial	
	The graph of the function f in the xy-plane crosses the x-axis at -4, 2 and 5. Which of the following could define f? A. $f(x) = (x-2)^2(x-5)$ B. $f(x) = (x^2+2x-8)(3x-15)$ C. $f(x) = (x^2-7x+10)(x-4)$ D. $f(x) = (x-4)(x+2)(x+5)$		
35	Advanced Math	Quadratic Function	
	 The graph of the function f in the xy-plane above is a parabola. Which of the following defines f? A. $f(x) = 4(x-3)^2 + 1$ B. $f(x) = 4(x+3)^2 + 1$ C. $f(x) = (x-3)^2 + 1$ D. $f(x) = 3(x-3)^2 + 1$		
36	Advanced Math	Polynomials	
	The expression of $\frac{1}{3}x^2 - 4$ can be written as $\frac{1}{3}(x-k)(x+k)$, where k is a positive constant. What is the value of k? A. 2 B. 6 C. $\sqrt{6}$ D. $\sqrt{12}$		
37	Advanced Math	Exponential Word Problem	
	Henry open bank account and earn 5% interest , compounded annually, with an initial deposit of P dollars. Henry calculates that if he does not make any withdrawals or deposit after the initial amount then the balance of account after 2 years will be $1,653.75 . What is the value, in dollars, of Henry's initial deposit?		

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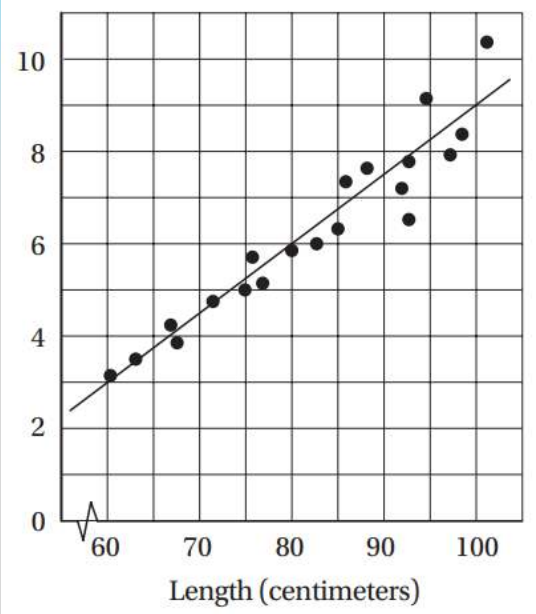
38	Problem Solving and Data Analysis	Scatterplot	



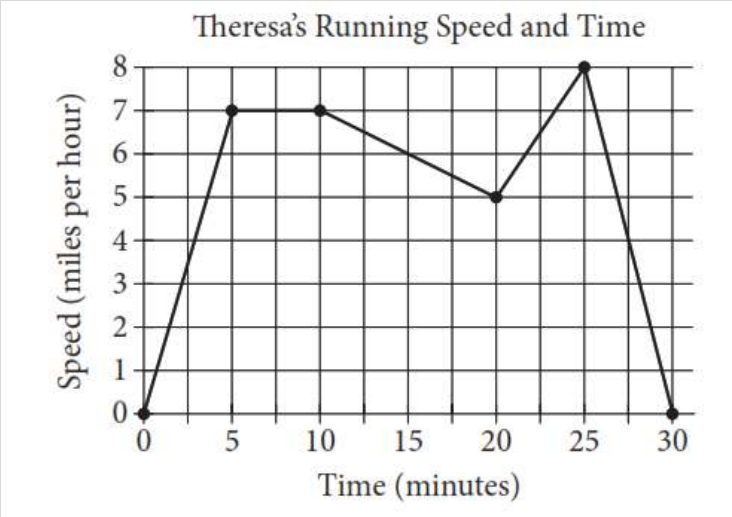
The scatterplot above shows the length and weight of a group of 20 tuna and the line of best fit for the data. According to this line of best fit, which of the following best approximates the weight, in kilograms, of a tuna that is 95 centimeters long?

A. 7.6
B. 7.8
C. 8.3
D. 8.6

39	Problem Solving and Data Analysis	Mean or Average	

A professor score 15 of 20 exams himself and gives the other 5 to his teaching assistant to score. If the average score of the 15 exams grade by the professor is 87 and the average of all 20 exams is 85, how many point lower the average score of the teaching assistant's 5 exam than the average score of the professor's 15 exams?

40	Problem Solving and Data Analysis	Graph	



Theresa ran on a treadmill for thirty minutes, and her time and speed are shown on the graph above. According to the graph, which of the following statements is true concerning Theresa's run?

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A) Theresa ran at a constant speed for 10 minutes.
B) Theresa's speed was increasing for a longer period of time than it was decreasing.
C) Theresa's speed decreased at a constant rate during the last five minutes.
D) Theresa's speed reached its maximum during the first ten minutes.

41 | Geometry | Circle Equation

Circle A has equation $(x - 5)^2 + (y + 2)^2 = 1$. In the xy – plane circle B is obtained by translating circle A to the left 5 units. What is the equation represent circle B?

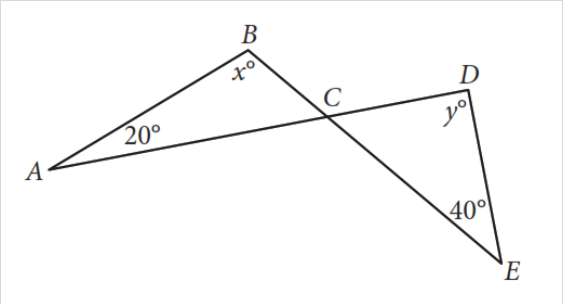
A. $(x - 10)^2 + (y + 2)^2 = 1$
B. $x^2 + (y + 2)^2 = 1$
C. $(x - 5)^2 + (y + 7)^2 = 1$
D. $x^2 + (y - 7)^2 = 1$

42 | Geometry | Area

If the length of a rectangle is decreased by 10 percent and its width is increased by p percent. It area will increased by 26 percent. What is the value of p?

A. 36
B. 38
C. 40
D. 42

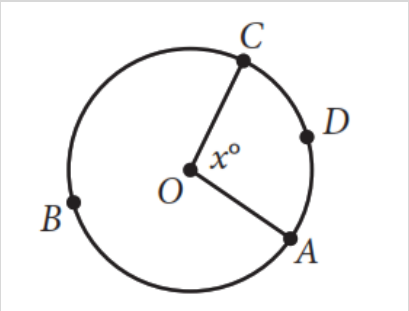
43 | Geometry | Line and Angle



In the figure above, AD intersect BE at C. If $y = 80$ what is the value of x?

A. 100
B. 90
C. 80
D. 60

44 | Geometry | Arcs of Circle



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	The circle above has center O. The length of arc ADC is 5π and the length of arc ABC is 13π. What is the value of x? A. 100 B. 130 C. 150 D. 260

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